



Characterization of radiation induced defects in EUROFER 97 after neutron irradiation

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ABSTRACT

Specimens of EUROFER 97 prepared for impact tests have been irradiated to an average dose of 16.3 dpa at irradiation temperatures of 250–450 °C. TEM investigations have been performed to study radiation induced changes in the microstructure. The characterization and statistical analysis show the temperature dependant formation of small dislocation loops and He bubbles. The Burgers vector of dislocation loops was $\frac{1}{2}\langle 111 \rangle$. A novel feature is that within statistical uncertainty the maximum in the dislocation density observed around 300 °C decreased with decreasing irradiation temperature down to 250 °C. The TEM data are correlated with tensile and instrumented Charpy test results.

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1. Introduction

The design, construction and reliable operation of environmentally attractive fusion power plants require highly radiation resistant reduced activation materials. The reduced activation ferritic martensitic (RAFM) steel 9%Cr–W–TaV, called EUROFER 97 has been systematically developed for this purpose in Europe [1–5]. Industrial batches of more than 20 tons have been produced by two different EU companies with a variety of different heats and semi-finished, quality-assured product forms (forged bars, plates, tubes and wires) in various dimensions. The database on these steels includes information from the qualification of unirradiated base and joined materials, and from a broad based neutron irradiation program. The later included mixed spectrum reactor irradiations of EUROFER 97 and EUROFER-type material in the HFR Petten reactor. Despite many favourable properties of RAFM 9%Cr–W–TaV steels, an intrinsic issue with FM steels is the irradiation induced hardening and embrittlement for irradiation temperatures below about 350–400 °C and specifically the pronounced tensile strain localization for irradiation and testing temperatures between about 330 and 400 °C.

This paper presents a TEM study of microstructural changes in EUROFER 97 specimens irradiated up to 16.3 dpa in the temperature range 250–450 °C. Special emphasis has been put on the temperature dependency of the irradiation induced loop density in order to get a better understanding of the recently observed temperature dependent hardening and embrittlement behavior.

Finally, the dislocation loop data might prove useful for the experimental validation of computer simulation results.

2. Experimental

Impact test specimens of EUROFER 97 were irradiated to an average dose of 16.3 dpa at irradiation temperatures of 250, 300, 350, 400 and 450 °C. The neutron irradiation was performed in the HFR (High Flux Reactor, Petten, Netherlands) at an average neutron flux of $3.99 \times 10^{18} \text{ m}^{-2} \text{ s}^{-1}$ ($E > 0.1 \text{ MeV}$). The TEM specimens were prepared and analyzed in the FML (Fusion Materials Laboratory) of the Hot Cell facility of Forschungszentrum Karlsruhe. The chemical composition of the EUROFER 97 heat (in wt.%) is: 0.12 C, 0.48 Mn, 1.08 W, 0.2 V, 0.14 Ta, 8.91 Cr, balance Fe.

Electrochemical thinning used a Tenupol-3 jet polisher with a 20% H_2SO_4 + 80% CH_3OH solution as electrolyte at a voltage of 10–12 V at 23 °C. Microstructure characterization used a FEI Tecnai 20 F TEM microscope suitable for analysing active specimens. The microscope was operated at 200 kV accelerating voltage with a field emission gun.

3. Results and discussion

3.1. Microstructural analysis

The bright and dark field TEM images, exhibiting small dislocation loops, of a EUROFER 97 specimen after irradiation at 250 °C is shown in the Fig. 1 as an example. Both images were obtained near [0 0 1] zone axis. The diameter of the dislocation loops or small radiation induced defects ranged from 2 nm to 15 nm, however the fraction of the 15 nm loops is low. In the case of small

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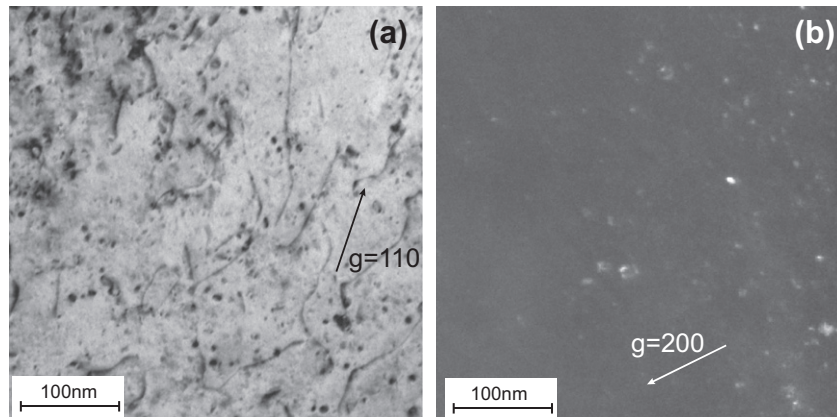


Fig. 1. TEM images of EUROFER 97 irradiated at 250 °C using $g = 1\ 1\ 0$ (a) and weak beam image $g(4g) g = 2\ 0\ 0$ (b).

Table 1

Statistical analysis of size and concentration of dislocation loops.

Irradiation temperature, °C	250	300	350	400	450
Defects concentration, cm^{-3}	2×10^{21}	4×10^{21}	3×10^{20}	5×10^{19}	1×10^{19}
Average size, nm	7	14	35	12	10

(2–5 nm) defects it is sometimes difficult to definitely determine whether they are small dislocation loops or point defect clusters. In the weak beam dark field image $g(4g) g = 0\ 2\ 0$ numerous small white dots and a few loops are visible (Fig. 1b). The concentration of the detected visible defects was estimated to be $2 \pm 0.5 \times 10^{21} \text{ m}^{-3}$ (Table 1). The dislocation loops have a $\frac{1}{2}(1\ 1\ 1)$ Burgers vector, determined using standard $g \cdot b = 0$ invisibility criteria. For this measurement the specimen was imaged near the $[0\ 0\ 1]$ zone axis.

Fig. 2 shows bright field images revealing dislocation loops in the specimens irradiated at 300 °C (a) and 350 °C (b). The irradiation induced defect density and defect size are significantly higher at 300 °C than at 250 °C. For irradiation at 300 °C, the total concentration of visible defects amounts to $4 \pm 0.8 \times 10^{21} \text{ m}^{-3}$. The diameter of dislocation loops is typically around 10 nm, only a few have grown to a diameter of 25 nm. Above an irradiation temperature of about 300 °C, the spatial concentration of irradiation induced defects strongly decreases, as shown in Fig. 2b. After irradiation at 350 °C dislocation loops with diameters up to 70 nm have been observed. The $\frac{1}{2}(1\ 1\ 1)$ Burgers vector was determined for all loops formed during neutron irradiation at 300 and 350 °C.

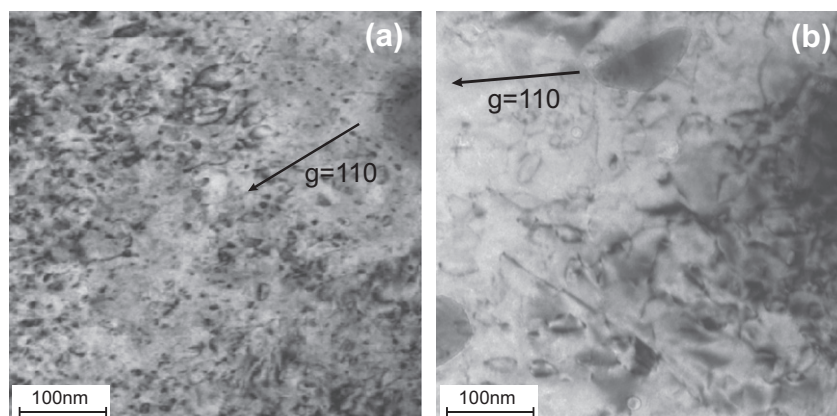


Fig. 2. TEM images of EUROFER 97 irradiated at 300 °C (a) and 350 °C (b) using $g = 1\ 1\ 0$ for the both images.

Fig. 3 shows TEM microstructures of specimens irradiated at 400 °C and 450 °C. Loops with diameters typically between 5 and 10 nm were detected, however the density of radiation induced defects is significantly smaller than for the lower temperature irradiations (Table 1).

In the specimens irradiated at 350 and 400 °C a small number density of voids or helium bubbles has been detected. Fig. 4 shows examples after 350 °C irradiation. Such voids or He bubbles are rare and many TEM micrographs do not contain any. However, if they show up, a couple of them can be detected within local areas of $200 \times 200 \text{ nm}$ (Fig. 4a or b). Due to the very low boron content of EUROFER 97, the total helium concentration is estimated as $\leq 10 \text{ appm}$ [5]. This inhomogeneous distribution of the rare cavities suggests that they are helium bubbles in the vicinity of boron rich precipitates. The density of bubbles in specimens irradiated at 400 °C is smaller than the specimen irradiated at 350 °C.

Table 1 presents the measured densities and average sizes of dislocation loops and He bubbles from the analysis of TEM micrographs. Only loops with $\frac{1}{2}(1\ 1\ 1)$ Burgers vector have been detected. This result correlates with literature where the formation of dislocations loops with $(1\ 0\ 0)$ Burges vector was observed only in specimens with lower Cr concentration and higher irradiation temperature [6].

3.2. Correlation of dislocation density with tensile and DBTT data

Tensile results from the same irradiated EUROFER 97 material reveals at $T_{\text{irr}} = 300 \text{ °C}$ an ultimate tensile strength R_m of $\sim 880 \text{ MPa}$ while at $T_{\text{irr}} = 250 \text{ °C}$ it is only $\sim 795 \text{ MPa}$ [5]. At first glance,

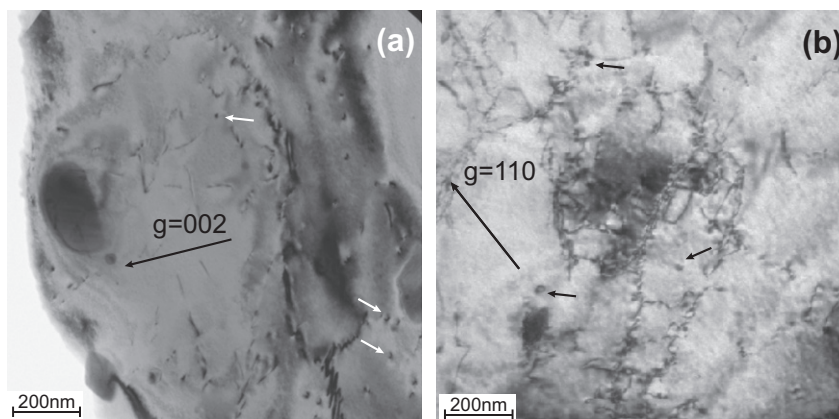


Fig. 3. TEM image of EUROFER 97 irradiated at 400 °C (a) using $g = 0\ 0\ 2$ and 450 °C (b) using $g = 1\ 1\ 0$. The dislocation loops are marked by the arrows.

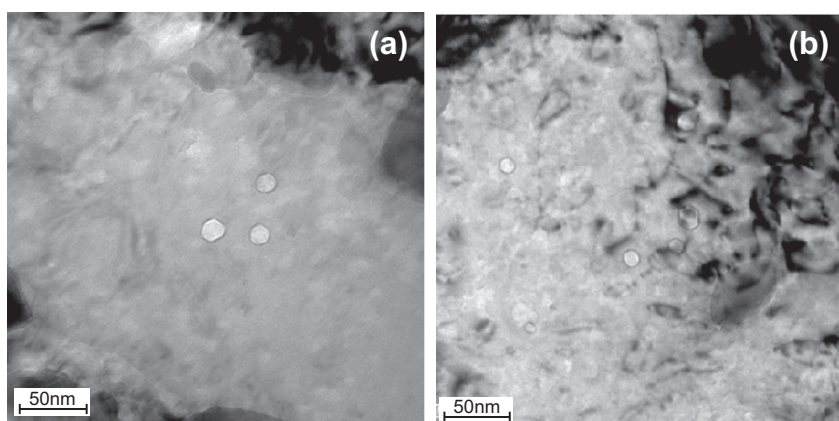


Fig. 4. Bright field TEM micrographs of He bubbles or voids in EUROFER 97 irradiated at 350 °C.

this behavior can be explained by dispersed barrier hardening model with a smaller irradiation induced defect density in the 250 °C specimens. The fact that with decreasing irradiation temperature the tensile strength does not show saturation around 300 °C but decreases again towards lower temperatures has been recently reported after an irradiation to 2.5 dpa [7]. The authors of that publication showed that 10 RAFM steels had a maximum R_m and a minimum total elongation at 300 °C. In instrumented Charpy tests, the ductile to brittle transition temperature (DBTT) seems to have a maximum for an irradiation temperature near 300 °C, although the data scatter is larger in this case.

Descriptions of the origin of the abnormal temperature dependency of the dislocation density and the relation to the structural properties based on diffusion kinetics and hardening mechanisms will be given in a forthcoming publication.

4. Summary

The density, size and Burgers' vector of irradiation induced defects in EUROFER 97 neutron irradiated to 16.3 dpa at 250–450 °C, have been analyzed by TEM. All loops had Burgers vector of $\frac{1}{2}(1\ 1\ 1)$. While the defect density revealed the expected strong de-

crease from 300 to 400 °C, an abnormal density decrease from 300 to 250 °C has been observed. However, this observation correlates with recent tensile and Charpy results that showed a maximum irradiation hardening and embrittlement around 300 °C. Very few and completely non-homogeneously distributed voids or helium bubbles have been found that can be attributed to the production of helium from segregated boron.

References

- [1] K. Ehrlich, W. Cierjacks, S. Kelzenberg, A. Möslang, The development of structural materials for reduced long-term activation, in: D. Gelles et al. (Ed.), 17th Int. Symposium on Effects of Radiation on Materials, ASTM STP 1270, 1996, pp. 1109–1122.
- [2] M.R. Gilbert, R.A. Forrest, Handbook of Activation Data Calculated Using EASY-2003, Report UKAEA FUS 509, July 2004.
- [3] B. van der Schaaf, F. Tavassoli, C. Fazio, E. Rigal, E. Diegele, R. Lindau, G. Le Marois, Fusion Eng. Des. 69 (2003) 197.
- [4] R. Lindau, A. Möslang, M. Schirra, Fusion Eng. Des. 61 and 62 (2002) 659.
- [5] E. Materna-Morris, A. Moeslang, R. Rolli, H.-C. Schneider, J. Nucl. Mater. 386–388 (2009) 422–425.
- [6] R. Schaublin, J. Henry, Y. Dai, C.R. Phys. 9 (2008) 389–400.
- [7] E. Materna-Morris, H.-C. Schneider, B. Dafferner, R. Rolli, O. Romer, A. Moeslang, in: SOFE2009, May 31–June 5, 2009, San Diego, California, IEEE Trans. Plasma Sci.